2025 IEEE 17th International Conference on Computational Intelligence and Communication Networks (CICN) | 979-8-3315-8733-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/CICN67655.2025.11367884

AI-Powered Automated System for Early Identification of Crop Diseases using Transfer Learning

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Abstract*—Early and accurate identification of crop diseases is essential for safeguarding agricultural yield, reducing economic loss, and supporting sustainable farming. This study presents an AI-powered automated system for early crop disease detection using transfer learning with the ResNet50 convolutional neural network. Leveraging the strengths of the ResNet50 model pretrained on ImageNet, the system was fine-tuned using the publicly available PlantVillage dataset, which contains thousands of labeled images of healthy and diseased crop leaves. The proposed approach achieved a remarkable 98.75% accuracy in classifying various plant diseases, including fungal, bacterial, and viral infections across multiple crop types. The use of transfer learning significantly reduced training time and computational cost while maintaining high generalization performance. Experimental results demonstrate the model's superiority over traditional machine learning techniques in both precision and scalability. This AI-driven solution can be integrated into precision agriculture tools, enabling real-time, field-level disease diagnosis to assist farmers in taking timely and informed actions for crop management.

Keywords*—Crop Disease, Crop growth tracking, Deep Learning, Automated model.

I. INTRODUCTION

Alongside technological progress, agriculture often seen as the foundation of any nation is transforming. Our survival depends heavily on crops [1]. However, one concentrate question bothering world today is food insecurity, which is a major contributing factor to crop diseases. In addition to endangering food security, plant diseases have a severe detrimental effect to livelihoods and economies. Therefore, it is crucial to keep an eye on crop health, which is mostly represented in the growth and condition of leaves, to guarantee both food safety and economic stability. Analyzing leaf symptoms is essential for identifying a variety of plant diseases [2]. In a few decades, the world's population is expected to reach 9 billion, indicating that food production must rise by 70% [3]. Addressing this issue is essential, particularly in agrarian countries, as crop diseases cause an annual production loss of 16%. Any illness that affects crops like potatoes, tomatoes, and peppers is concerning because of their widespread use. The greatest strategy is still prevention.

Machine learning approaches have shown promise in day to day activities and data extraction as computing has advanced. Deep learning has been investigated by agricultural scientists as a potential weapon against potato crop diseases. In particular, deep learning is essential for resolving challenging issues in agriculture, encompassing image categorization, pattern recognition, and feature extraction [4-8]. In order to minimize yield loss by real-time deployment, this study suggests a CNN-based method for crop disease diagnosis. Rural farmers frequently lack access to agricultural experts and must rely on their experience or local expertise, which can be costly, time-consuming. AI-based solutions can minimize crop damage and greatly speed up the detecting process.

AI based models gives best alternatives for automated crop disease prediction with high exactness and proficiency [11]. The Inception-ResNet-v2 model can accurately identify crop diseases with an overall recognition accuracy of 86.1% [12]. Deep learning-based solutions can accurately identify and detect insects in soybean crops with 98.75%, 97%, and 97% accuracy, respectively [13]. The proposed model accurately classifies disease-affected leaves in tomatoes and grapes with a 98.40% accuracy [14]. AI algorithms can rapidly and accurately identify various crop diseases, enhancing agricultural productivity and global food security [15]. Automated leaf disease diagnosis using AI and IoT

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DOI: 10.1109/CICN.2025.295

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sensors can enhance agricultural yield by predicting plant diseases in tomatoes, chillies, potatoes, and cucumbers [16]. An ensemble of lightweight LSTM model for plant disease detection in IoT devices achieves high classification accuracy and low prediction time with fewer trainable parameters [17]. The YOLOv8n deep convolutional neural network successfully detects and classifies tomato diseases with 96.5 % accuracy, 97.5% precision, and 93.8 % recall [18]. YOLOv8 nano, an advanced machine learning model, offers a mean average precision of 98.6% for early detection of plant diseases in tomato crops [19]. Two AI models, Inception V3 and ResNet-9, can detect and classify various crop diseases using Deep Learning, significantly increasing productivity and yield [20].

By utilizing satellite images, IoT sensors positioned in fields, as well as historical records of pest and disease incidences, a forecast monitoring system center that works based on the data's underlying algorithms, is able to promptly issue alerts concerning potential crop diseases and pest infestations which enables better within season intervention providing early warning systems that aid in effective agricultural pest control while cultivating sustainable agricultural practices and enhancing food security [21]. Such timely information systems can also be designed to automate crop disease detection at an earlier stage through the use of image processing techniques in combination with machine learning or deep learning frameworks allowing for precision and minimal crop loss through early targeted interventions [22]. Many scholar attempted to develop an automated classification models [23-28]. These techniques helped in detecting illnesses in tomatoes cultivated in greenhouses and fields.

The motto of this study is to detect the abnormal leaves or leaves with virus and fungi and start pesticides to prevent damage of crops.

II. METHODOLOGY

Fig.1 shows the methodology of our proposed approach. This methodology is designed to detect diseases on the leaf real-time deep-learning (DL) algorithms. Pre-trained DL network-ResNet is trained and tested using images taken from the PlantVillage database, which covered ten classes-including that of healthy leaves internet images for testing. Once the images are collected, they undergo various steps to upgrade the model's precision and efficiency.

There are various steps that are involved in developing a DL model. The first and foremost step is preprocessing. Preprocessing ensures that the training data is in the correct format for training. In preprocessing step, images are rotated, resized and rescaled into proper format that is adaptable by the DL models used. The next step is to do data augmentation which increases the dataset size and enhances the generalization capability. The augmented dataset is now divided in two sections-Training data and testing data. While training the model goes through multiple training epochs, in which it modifies weights according to the data in order to decrease error. Gradient descent and back propagation techniques are applied. The test data is then applied to evaluate the performance of this model in identifying and predicting crop diseases. It includes measures like accuracy, precision, recall, and F1-score for evaluation. After training and validating the model, it is optimized with different optimizers to get the best outcome. The developed model can further be deployed for various applications. The system can analyze new leaf images uploaded by users and provide disease predictions.

A. Dataset

In this research, crop images of are taken from Plant Village Dataset. Table 1 shows the leaf diseases considered and the notation given for each class. Table 2 gives the summary of images taken from dataset. Fig.2 displays sample image of each leaf class.

TABLE 1. AFFECTED PLANTS CONSIDERED FROM PLANTVILLAGE DATASET

Class	Disease	Affected Plant
D1	Healthy	Tomato, Potato, Black Pepper
D2	Bacteria spot	Tomato, Black Pepper
D3	Spider mites	Tomato
D4	Late blight mold	Tomato, Potato
D5	Virus Leaf	Tomato, Black Pepper
D6	Fungus leaf	Tomato, Potato

TABLE 2. SUMMARY OF IMAGES TAKEN FROM PLANTVILLAGE DATASET

Class	Total data	Training data	Validation data	Testing data
D1	4909	3139	788	982
D2	4337	2776	694	867
D3	1341	859	214	268
D4	2327	1489	372	466
D5	4585	2935	733	917
D6	4101	2625	656	820
Total	**21600**	**13823**	**3457**	**4320**

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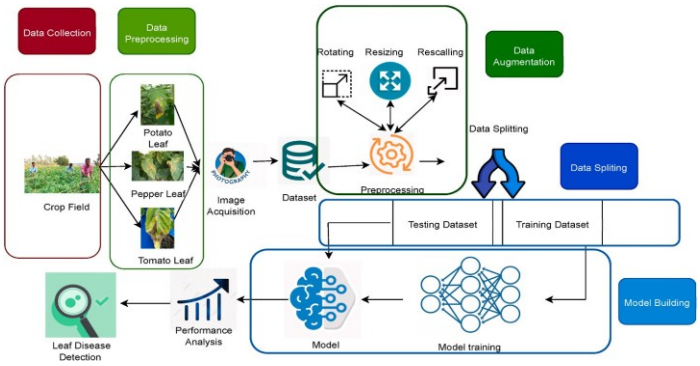
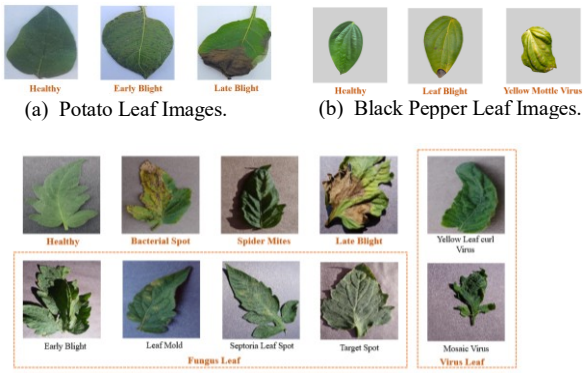


Fig.1 Proposed Workflow Diagram



(a) Potato Leaf Images. (b) Black Pepper Leaf Images.

(c) Tomato Leaf Image.

Fig.2 Sample Images from Plant Village Dataset

B. Deep Learning(DL) model

DL architectures: ResNet50 is used in this study. ResNet-50 is a 50-layer deep CNN that is popular for using residual connections to ease the problem caused due to decreasing gradient. It was first introduced in 2015 and won the Image Net competition. ResNet-50 is a computationally efficient and accurate models that is widely used for image classification applications. It is composed of residual blocks, followed by global normal pooling and a fully associated layer for classification. It is popular in applications like object detection, segmentation, and feature extraction given it strikes a good balance between model difficulty and computational cost, making it appropriate for both research and production settings. Fig. 3 shows the pretrained ResNet50 architecture.

III. RESULTS AND DISCUSSIONS

The research work is carried on system with provisions-Intel Core i9-10th Gen, with 32 GB RAM, and NVIDIA GeForce GTX 1060 4GB Graphic card. Training hyper-parametrs selected are shown in Table 1.

The hyperparameters presented in Table 1 significantly influence the training dynamics and overall performance of the proposed ResNet50-based crop disease detection model. A total of 20 training epochs were used to allow the model adequate exposure to the dataset while minimizing the risk of overfitting. The learning rate was set to a very small value of 0.00001 to ensure stable and gradual weight updates, which is particularly important when fine-tuning a pretrained deep network like ResNet50. A batch size of 10 was chosen to allow frequent weight updates, which can lead to faster convergence, especially when working with limited computational resources. The RMSprop optimizer was employed due to its ability to adaptively adjust learning rates for each parameter, making it suitable for image classification tasks involving noisy gradients. The multiclass entropy loss function (categorical cross-entropy) was used to effectively measure the error in multi-class classification, guiding the optimization process during training. Finally, the SoftMax activation function in the output layer ensures that the model's predictions are normalized into a probability distribution across the various disease classes. Collectively, these hyperparameter choices contributed to the model's high accuracy of 98.75% in detecting plant diseases using the PlantVillage dataset. Training graphs of developed models is shown in Fig.3,Fig.4.

TABLE 1. HYPERPARAMETERS STUDIED

Training Parameters	**Value**
Training Epoch	20
Learn Rate	0.00001
Batch Size	10
Optimizer	RMS prop
Loss Function	Multi class entropy loss function
Activation function	SoftMax

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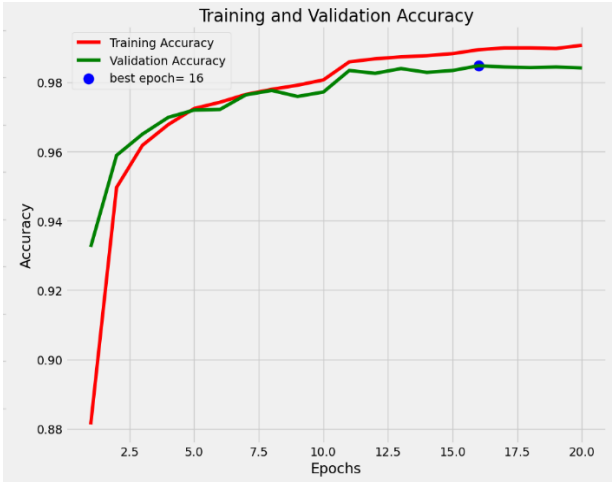


Fig.3 Training and Validation Accuracy Graph

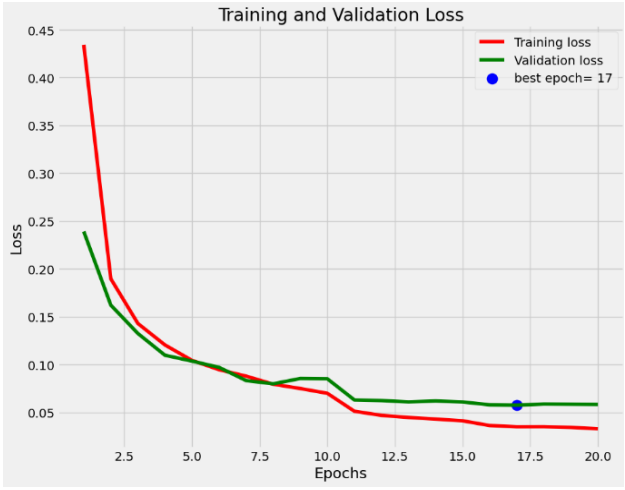


Fig.4 Training and Validation Loss Graph

The confusion matrix obtained is shown in Fig.5. True labels are considered on X-axis and Predicted labels are considered on Y-axis. To efficacy of developed DL models is assessed by performance metrics like accuracy(A), precision(P), recall(R), and F-score(F). Comparison of measures for each class is shown in Fig.6.

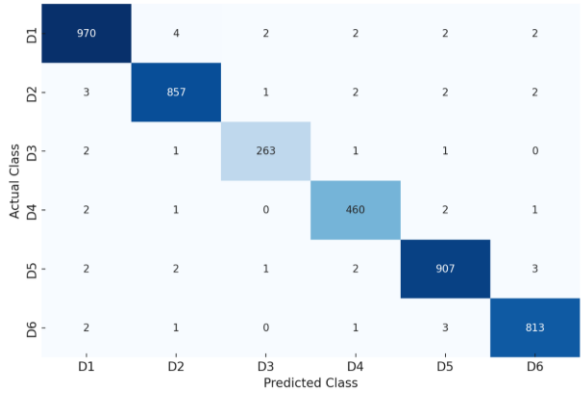


Fig.5 Confusion chart

TABLE 5. PERFORMANCE MEASURES OBTAINED USING RESNET50

	Precision	Sensitivity	Specificity	F-score
D1	0.9888	0.9878	0.9967	0.9883
D2	0.9896	0.9885	0.9974	0.9890
D3	0.9850	0.9813	0.9990	0.9832
D4	0.9829	0.9871	0.9979	0.9850
D5	0.9891	0.9891	0.9971	0.9891
D6	0.9903	0.9915	0.9977	0.9909
Accuracy= 98.75 %				

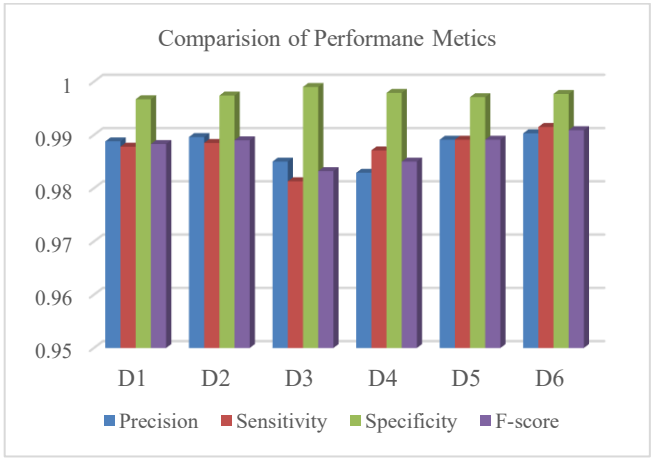


Fig.6 Comparison of performance of each class

These obtained metrics confirm that the model performs consistently well across all disease classes, with F-scores close to or above 0.98, indicating excellent precision-recall balance. Table 5 shows the resemblance in accuracy between our DL algorithm and the state of the art methods currently used.

TABLE 5. COMPARISONS OF STATE-OF-ART MODELS

Study	Method	Accuracy (%)
S.Ashok et al [23]	CNN	98.12 %
M. Agarwal et al [24]	CNN	91.2 %
Wang et al [35]	CNN	97 %
Cheemaladinne et al [26]	CNN+GAN	92 %
Agarwal et al [27]	VGG16	91.2 %
Proposed Approach	Automated CNN	98.75 %

IV. CONCLUSION

Detecting and classifying leaf diseases is a crucial yet challenging issue in agriculture globally. To identify leaf illnesses and their causes, this study made use of CNN architectures. Extensive studies were performed with the "Plant Village" dataset, which is freely available. For ResNet50 DL model accuracy rates of 98.75% was observed. Therefore, to accurately identify plant leaf diseases, we implemented the model with highest precision in our online application. The online application gives a clever agricultural

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disease detection solution. The suggested approach gives great results for valuing symptom severity equated to previous studies.

In the future, some plans to enhance the accuracy to develop AI-based techniques of incorporating an attention mechanism and implement a local technique at the affected area on the leaf. Additionally, we also aim to utilize more diverse and comprehensive multi-leaf disease datasets to cover a wide range of plant leaf diseases.

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